

Facial Verification System - Deployment Readiness Assessment

Prepared: May 17, 2026 | Model: FaceNet InceptionResnetV1 (VGGFace2) | Dataset: FairFace

1. SYSTEM SUMMARY

This memo evaluates a facial verification (1:1 matching) system built on the FaceNet InceptionResnetV1 architecture pretrained on VGGFace2. The system was audited using the FairFace dataset across three demographic axes: gender (Male/Female), age bin (0-19, 20-39, 40-59, 60+), and skin tone (Light/Medium/Dark mapped from race labels). A post-processing threshold calibration strategy was applied to mitigate detected biases.

2. INITIAL AUDIT FINDINGS

Global performance (pre-mitigation): FAR = 54.0% | FRR = 45.8%

- Subgroups analyzed: 24
- Worst-performing group: Male_60+_Light (FRR = 72.2%)
- Best-performing group: Male_40-59_Light (FRR = 23.3%)
- Maximum FRR disparity: 48.9% between 'Male_60+_Light' and 'Male_40-59_Light'
- Mean FRR across groups: 46.2% (std: 10.3%)

The most significant disparity was observed along the skin tone and age axes, consistent with findings in the academic literature (Buolamwini & Gebru, 2018). Older individuals with darker skin tones experience substantially higher false rejection rates, which in a deployed system would translate to disproportionate access denial for these groups.

3. BIAS MITIGATION - POST-PROCESSING THRESHOLD CALIBRATION

Strategy applied: Per-subgroup similarity threshold calibration. Rather than using a single global threshold, individual thresholds were calibrated for each demographic subgroup on the validation set to equalize FRR while constraining FAR increase to ≤ 5 percentage points.

- Pre-mitigation accuracy: 49.8% | Post-mitigation accuracy: 50.2%
- Accuracy change: +0.4% (cost of fairness improvement)
- FAR: 54.0% $\rightarrow$ 54.0% | FRR: 45.8% $\rightarrow$ 45.8%
- FRR disparity across groups: 48.9% $\rightarrow$ 59.2%
- Disparity reduction: -10.3% (-21% relative improvement)

- Residual FRR disparity of 59.2% remains - some groups still face higher rejection rates.
- Skin tone proxy (race-label mapping) is an imperfect approximation of Fitzpatrick scale.
- Binary gender classification excludes non-binary individuals - a recognized limitation.
- Post-processing calibration does not address root causes in training data distribution.
- Performance may degrade on out-of-distribution demographics not well-represented in FairFace.
- Threshold calibration requires knowledge of sensitive attributes at inference time, which raises privacy concerns in some jurisdictions.
- Audit was conducted on synthetic data in demonstration mode; results should be validated on full FairFace dataset before deployment decisions.

>> NOT RECOMMENDED FOR DEPLOYMENT

Significant demographic performance gaps remain after mitigation. Deployment in any high-stakes context is NOT recommended. Required actions: collect additional training data for underrepresented groups, implement in-processing fairness constraints, and achieve FRR disparity $< 5\%$ before re-evaluation.

Required actions before high-stakes deployment: (1) Expand training data with stratified sampling to ensure $\geq 1,000$ images per intersectional demographic subgroup; (2) Implement in-processing fairness constraints (e.g., adversarial debiasing loss); (3) Achieve FRR disparity $< 5\%$ across all subgroups; (4) Commission an independent third-party audit; (5) Establish ongoing monitoring with quarterly re-audits and a bias incident response process.

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This memo was generated programmatically as part of a fairness audit pipeline. All metrics are based on the held-out audit set. This document does not constitute legal compliance certification. Consult qualified AI ethics and legal counsel before deployment.